

Quantum Simulation of Spin-Dependent Charmonium Spectroscopy with Variational Quantum Algorithms

Master's Thesis

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Chapter 1

Introduction

The simulation of quantum systems lies at the heart of quantum computing’s promise. As Feynman observed in 1982, classical computers face a fundamental barrier when modelling quantum mechanical systems: the Hilbert space of an n -particle system grows exponentially, rendering exact classical computation intractable for even modest system sizes [1]. Quantum computers, by contrast, naturally encode and manipulate quantum states, offering a path toward efficient simulation of physical systems that are otherwise inaccessible [2].

This thesis applies variational quantum algorithms to the spectroscopy of charmonium—bound states of a charm quark and its antiquark—including spin-dependent interactions that produce the experimentally observed mass splittings between states such as the η_c and J/ψ mesons. The quarkonium system is physically rich enough to probe the capabilities of near-term quantum algorithms across multiple quantum number channels and excited states, yet small enough to be tractable with only a few qubits.

The remainder of this chapter introduces the three pillars on which this work rests: the fundamentals of quantum computing (Section 1.1), variational quantum methods (Section 1.2), and the quarkonium problem (Section 1.3).

1.1 Quantum Computing

1.1.1 Why quantum computing?

The classical theory of computation, formalized by Turing and Church in the 1930s, rests on the assumption that information is processed according to classical physics. For decades this abstraction posed no practical limitation. However, the simulation of quantum mechanical systems reveals a fundamental barrier: the state space of an n -particle quantum system is a Hilbert space of dimension 2^n , so that a faithful classical representation requires resources that grow exponentially with system size. A modest system of 50 interacting spin- $\frac{1}{2}$ particles, for instance, demands $2^{50} \approx 10^{15}$ complex amplitudes—

exceeding the memory of any existing classical computer.

In 1982, Feynman proposed a radical alternative: rather than simulating quantum mechanics on a classical machine, one should use a controllable quantum system to simulate another [1]. The exponential scaling that makes classical simulation intractable becomes an *asset*: a quantum processor with n qubits naturally inhabits the same 2^n -dimensional space as the target system. This observation laid the conceptual foundation for quantum computing.

The subsequent development of quantum algorithms by Deutsch, Shor, and Grover [2] demonstrated that quantum computers are not merely faster classical machines, but represent a genuinely different model of computation. The complexity class BQP (bounded-error quantum polynomial time) is believed to be strictly larger than its classical counterpart BPP, meaning there exist problems for which quantum algorithms achieve an exponential advantage that no classical algorithm can match [2]. The source of this advantage lies in two uniquely quantum phenomena: *superposition*, which allows exploration of exponentially many computational paths simultaneously, and *interference*, which enables the selective amplification of correct answers and suppression of incorrect ones.

1.1.2 Qubits and quantum states

The fundamental unit of quantum information is the *qubit*, a two-level quantum system whose pure state is described by a normalized vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1.1)$$

where $\alpha, \beta \in \mathbb{C}$ are *probability amplitudes* and $\{|0\rangle, |1\rangle\}$ is the computational basis (conventionally identified with the eigenstates of the Pauli Z operator). Unlike a classical bit, which is deterministically 0 or 1, a qubit exists in a coherent *superposition* of both basis states.

Geometrically, any single-qubit pure state can be represented as a point on the *Bloch sphere* via the parameterization

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (1.2)$$

with polar angle $\theta \in [0, \pi]$ and azimuthal angle $\phi \in [0, 2\pi)$. The north and south poles correspond to $|0\rangle$ and $|1\rangle$, respectively, while points on the equator represent equal superpositions with varying relative phase.

For a system of n qubits, the state space is the tensor product $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, spanned by 2^n computational basis states $|x_1 x_2 \cdots x_n\rangle$ with $x_i \in \{0, 1\}$. A general pure state is

therefore

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} c_x |x\rangle, \quad \sum_x |c_x|^2 = 1, \quad (1.3)$$

requiring 2^n complex amplitudes to specify. It is this exponential growth of the state space that makes quantum systems both hard to simulate classically and powerful as computational devices.

1.1.3 Entanglement

A multi-qubit state is called *separable* (or a product state) if it can be written as a tensor product of individual qubit states, $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$. States that *cannot* be decomposed in this way are *entangled*. The simplest example is the Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad (1.4)$$

which describes two qubits in a joint superposition: neither qubit has a definite value, yet their measurement outcomes are perfectly correlated. This correlation persists regardless of the spatial separation between the qubits and cannot be explained by any local hidden variable theory, as demonstrated by Bell's theorem [3].

Entanglement is not merely a theoretical curiosity—it is a computational resource. Classical correlations can be shared between bits using shared randomness, but entangled qubits carry correlations that are strictly stronger. In quantum algorithms, entangling gates create superpositions over correlated multi-qubit configurations, enabling the exploration of an exponentially large solution space in a way that has no classical counterpart. Without entanglement, a quantum computer can be efficiently simulated classically [2]; it is entanglement that unlocks the full power of quantum computation.

1.1.4 Quantum gates and circuits

The time evolution of a closed quantum system is governed by a unitary operator. Quantum computation exploits this by decomposing the desired evolution into a sequence of elementary *quantum gates*: unitary transformations acting on one or two qubits at a time.

The three Pauli operators form a basis for single-qubit operations:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.5)$$

Together with the identity I , they satisfy $\sigma_i^2 = I$, anticommute pairwise ($\sigma_i \sigma_j = -\sigma_j \sigma_i$ for $i \neq j$), and form a basis for the space of 2×2 Hermitian matrices. Any 2×2 Hermitian operator—and hence any single-qubit observable—can be written as $A = a_0 I + a_1 X + a_2 Y + a_3 Z$ with $a_i \in \mathbb{R}$.

The Pauli operators generate continuous families of rotation gates on the Bloch sphere:

$$R_\mu(\theta) = e^{-i\theta\mu/2} = \cos\frac{\theta}{2} I - i \sin\frac{\theta}{2} \mu, \quad \mu \in \{X, Y, Z\}. \quad (1.6)$$

The Hadamard gate $H = (X + Z)/\sqrt{2}$ maps $|0\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$, creating an equal superposition from a computational basis state.

Two-qubit entangling gates introduce correlations between qubits. The controlled-NOT (CNOT) gate flips the target qubit conditioned on the control qubit being in state $|1\rangle$:

$$\text{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (1.7)$$

The set $\{R_X, R_Y, R_Z, \text{CNOT}\}$ constitutes a *universal gate set*: any unitary on n qubits can be approximated to arbitrary precision by a finite sequence of these gates [2].

A quantum algorithm is specified as a *quantum circuit*: a sequence of gates applied to a register of qubits initialized in a known state, followed by measurement. Figure 1.1 illustrates a minimal circuit that prepares the Bell state of Eq. (1.4) using only a Hadamard gate and a CNOT.

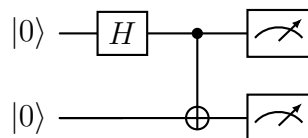


Figure 1.1: Quantum circuit preparing the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. The Hadamard gate creates a superposition on the first qubit; the CNOT entangles both qubits. Measurement yields $|00\rangle$ or $|11\rangle$, each with probability $1/2$.

1.1.5 Quantum parallelism and interference

The mechanism behind quantum computational advantage is often described loosely as “computing on all inputs at once.” While evocative, this picture is incomplete: the key ingredient is not parallelism alone, but the ability to make the parallel branches *interfere*.

Consider a unitary U_f that evaluates a function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. Applied to an equal superposition over all 2^n inputs, it produces

$$U_f \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |f(x)\rangle. \quad (1.8)$$

All function values are encoded in a single quantum state. However, measuring this state yields a *single* random pair $(x, f(x))$ —no better than classical evaluation. The power of

quantum algorithms lies in subsequent operations that exploit *interference*: unitary gates are designed so that amplitudes corresponding to desired outputs add constructively, while those corresponding to undesired outputs cancel destructively. The final measurement then yields the correct answer with high probability.

This interplay of superposition, entanglement, and interference is the mathematical core of quantum speedups. Superposition provides exponential parallelism; entanglement enables correlations across qubits that have no classical equivalent; and interference channels the computation toward the correct outcome.

1.1.6 Measurement and the Born rule

While unitary evolution is deterministic, extracting information from a quantum system is inherently probabilistic. A *projective measurement* in the computational basis is described by the set of projectors $\{\Pi_x = |x\rangle\langle x|\}_{x \in \{0,1\}^n}$. Given a state $|\Psi\rangle = \sum_x c_x |x\rangle$, the *Born rule* prescribes the outcome probabilities:

$$p(x) = |\langle x|\Psi\rangle|^2 = |c_x|^2. \quad (1.9)$$

Upon obtaining outcome x , the state collapses to $|x\rangle$, and any information encoded in the remaining amplitudes is irretrievably lost.

This has a direct practical consequence: a single measurement of n qubits yields only n classical bits, despite the state containing 2^n amplitudes. To estimate a continuous quantity—such as an energy expectation value—the measurement must be repeated many times (N_{shots} repetitions), and the result is subject to statistical uncertainty scaling as $1/\sqrt{N_{\text{shots}}}$.

More generally, to measure an observable O (a Hermitian operator), its expectation value in state $|\Psi\rangle$ is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle = \sum_i \lambda_i p_i, \quad (1.10)$$

where λ_i are the eigenvalues of O and p_i the probability of obtaining the corresponding eigenstate. When O is decomposed as a sum of Pauli operators, each Pauli term can be measured independently by rotating to the appropriate eigenbasis before measurement in the computational basis.

1.1.7 Density matrices and mixed states

The pure state formalism of Eq. (1.1) describes isolated, perfectly controlled quantum systems. Real quantum processors, however, interact with their environment, leading to *decoherence* and *mixed states* that require a more general description.

The *density matrix* (or density operator) of a pure state $|\psi\rangle$ is

$$\rho = |\psi\rangle\langle\psi|. \quad (1.11)$$

A *mixed state*—representing classical uncertainty about which pure state the system is in—is described by a convex combination

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1. \quad (1.12)$$

The density matrix is Hermitian, positive semidefinite, and has unit trace. A state is pure if and only if $\text{Tr}(\rho^2) = 1$; for mixed states, $\text{Tr}(\rho^2) < 1$, quantifying the degree of decoherence.

In the density matrix formalism, unitary evolution becomes $\rho \mapsto U\rho U^\dagger$, and expectation values are computed as $\langle O \rangle = \text{Tr}(O\rho)$. Noise processes—such as the depolarizing channel relevant to this thesis—are described by completely positive trace-preserving (CPTP) maps on ρ . For example, the single-qubit depolarizing channel with error rate p acts as

$$\mathcal{E}(\rho) = (1 - p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z), \quad (1.13)$$

which with probability p replaces the state with a random Pauli error. This formalism will be central to the analysis of noise effects in Chapter 1 and subsequent chapters.

1.1.8 Quantum simulation

Among the many proposed applications of quantum computing—including cryptanalysis, optimization, and machine learning—the simulation of quantum systems stands out as the most natural and arguably the most promising near-term application [1, 4]. The reason is both conceptual and practical: a quantum computer *is* a quantum system, so using it to simulate another quantum system leverages its native capabilities rather than working against them.

The essential mapping is straightforward. A physical system with a 2^n -dimensional Hilbert space is encoded on n qubits. The system’s Hamiltonian H is decomposed into a sum of Pauli operators acting on these qubits:

$$H = \sum_{i=1}^M c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n}, \quad (1.14)$$

where each P_i is a tensor product of single-qubit Pauli matrices (a *Pauli string*) and $c_i \in \mathbb{R}$ are real coefficients. This decomposition is always possible since the 4^n Pauli strings form a complete basis for $2^n \times 2^n$ Hermitian matrices.

Ground state energies, excited state spectra, and dynamical properties can then be

extracted using quantum algorithms that prepare and measure appropriate states on the quantum register. For the small systems considered in this thesis, the encoding requires only 2–3 qubits, but the same framework scales to larger systems where classical methods become intractable.

Quantum simulation of atomic, molecular, and nuclear systems has seen rapid progress in recent years, with demonstrations ranging from small molecules to lattice gauge theories [5]. The quarkonium system studied in this thesis belongs to this broader programme, representing a step toward quantum simulation of QCD bound states.

1.1.9 The NISQ era

Current quantum hardware operates in the *Noisy Intermediate-Scale Quantum* (NISQ) regime [4], characterized by processors with tens to hundreds of qubits subject to significant imperfections:

- **Gate errors** on the order of 10^{-3} – 10^{-2} per operation, described by noise channels such as the depolarizing channel of Eq. (1.13);
- **Limited connectivity** between qubits, requiring additional SWAP gates that increase circuit depth;
- **Short coherence times** that bound the number of operations executable before decoherence degrades the quantum state.

These constraints preclude the deep circuits required by fault-tolerant algorithms such as quantum phase estimation, which need thousands of controlled operations and ancilla qubits for error correction.

The NISQ era has therefore driven the development of algorithms specifically designed for shallow circuits and resilient to moderate noise levels. The most successful paradigm to date is the *variational* approach, in which a quantum processor prepares parameterized trial states while a classical computer optimizes the parameters—a hybrid strategy discussed in the next section.

1.2 Variational Quantum Methods

1.2.1 The variational principle

Variational methods offer a framework for extracting useful results from NISQ hardware. The central idea rests on the *Rayleigh–Ritz variational principle*: for a Hamiltonian H with ground state energy E_0 ,

$$E_0 \leq \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle \equiv E(\boldsymbol{\theta}) \quad (1.15)$$

for any normalized trial state $|\psi(\boldsymbol{\theta})\rangle$ parameterized by classical variables $\boldsymbol{\theta} \in \mathbb{R}^p$. Minimizing $E(\boldsymbol{\theta})$ over the parameter space yields an upper bound on E_0 , with equality when $|\psi(\boldsymbol{\theta})\rangle$ includes the true ground state in its span.

1.2.2 The hybrid quantum–classical loop

In the quantum computing implementation [5, 6], the trial state is prepared by a *parameterized quantum circuit* (also called an *ansatz*) applied to an initial reference state:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle^{\otimes n}, \quad (1.16)$$

where $U(\boldsymbol{\theta})$ is a product of parameterized and fixed gates. The energy expectation value is then estimated on the quantum processor. For a Hamiltonian decomposed as a weighted sum of Pauli strings,

$$H = \sum_{i=1}^M c_i P_i, \quad P_i \in \{I, X, Y, Z\}^{\otimes n}, \quad (1.17)$$

each expectation value $\langle P_i \rangle$ is estimated independently through repeated state preparation and measurement in the appropriate Pauli basis. The total energy,

$$E(\boldsymbol{\theta}) = \sum_{i=1}^M c_i \langle P_i \rangle_{\boldsymbol{\theta}}, \quad (1.18)$$

is passed to a *classical optimizer* that updates $\boldsymbol{\theta}$ to minimize E . This loop repeats until convergence, as illustrated in Fig. 1.2.

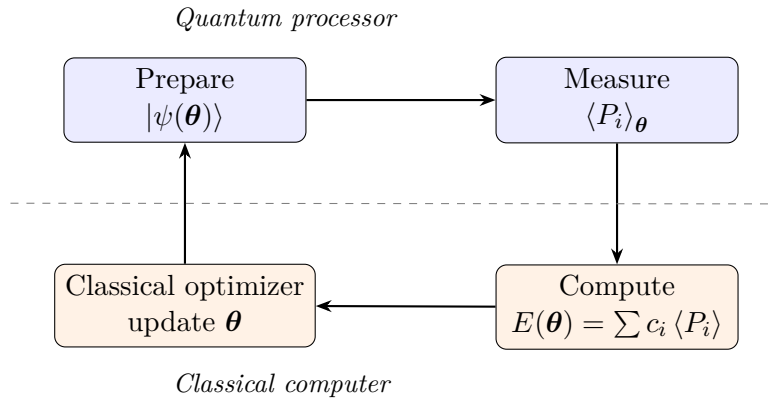


Figure 1.2: Schematic of the hybrid quantum–classical variational loop. The quantum processor prepares a parameterized state and estimates Pauli expectation values; the classical computer evaluates the energy and updates circuit parameters. The loop iterates until convergence.

1.2.3 Ansatz design and practical considerations

The choice of ansatz circuit is problem-dependent and critical to performance. *Hardware-efficient* ansätze use gates native to the target device, minimizing circuit depth at the cost of physical motivation. *Physically-motivated* ansätze, such as those inspired by unitary coupled cluster theory from quantum chemistry, encode domain knowledge into the circuit structure and may converge faster toward the physical ground state.

The hybrid approach offers several advantages for near-term hardware. The quantum processor handles the exponentially hard task of preparing and measuring many-body quantum states, while the classical optimizer navigates the energy landscape using well-established numerical methods. Circuit depths remain manageable, and variational methods exhibit a degree of inherent noise resilience: small perturbations in the energy surface due to hardware errors are partially absorbed by the optimization [6].

Beyond the ground state, variational frameworks extend naturally to excited states through orthogonality constraints and penalty methods, and to quantum dynamics through variational formulations of time evolution. These extensions broaden the scope of problems addressable on near-term quantum hardware.

1.3 The Quarkonium Problem

1.3.1 Heavy quark bound states

Quantum chromodynamics (QCD), the gauge theory of the strong interaction, governs the binding of quarks and gluons into hadrons. Among the simplest hadronic systems are *quarkonia*: bound states of a heavy quark and its antiquark ($q\bar{q}$). Charmonium ($c\bar{c}$) systems have served as testing grounds for QCD since the simultaneous discovery of the J/ψ meson in 1974 at BNL [7] and SLAC [8]—an event known as the *November Revolution* in particle physics.

The large mass of the charm quark ($m_c \approx 1.27 \text{ GeV}/c^2$) implies that the quark–antiquark pair moves non-relativistically inside the bound state, with typical velocities $v/c \sim 0.3$. This justifies a potential model approach [9] in which the dynamics are governed by the non-relativistic Schrödinger equation with an effective interquark potential.

1.3.2 The Cornell potential

The *Cornell potential* [9],

$$V(r) = -\frac{\kappa}{r} + \sigma r, \quad (1.19)$$

provides a successful phenomenological description of the $q\bar{q}$ interaction. It captures the two essential features of the strong force:

- **Short-range Coulomb attraction** ($-\kappa/r$), arising from one-gluon exchange between the color charges, analogous to the electromagnetic Coulomb potential but with QCD coupling;
- **Long-range linear confinement** (σr), a consequence of the non-Abelian nature of QCD, modelled by a chromoelectric flux tube (string) with tension σ connecting the quark and antiquark.

Typical parameter values for charmonium are $\kappa \approx 0.52$ and $\sigma \approx 0.183 \text{ GeV}^2$ [9]. Figure 1.3 shows the characteristic shape of this potential.

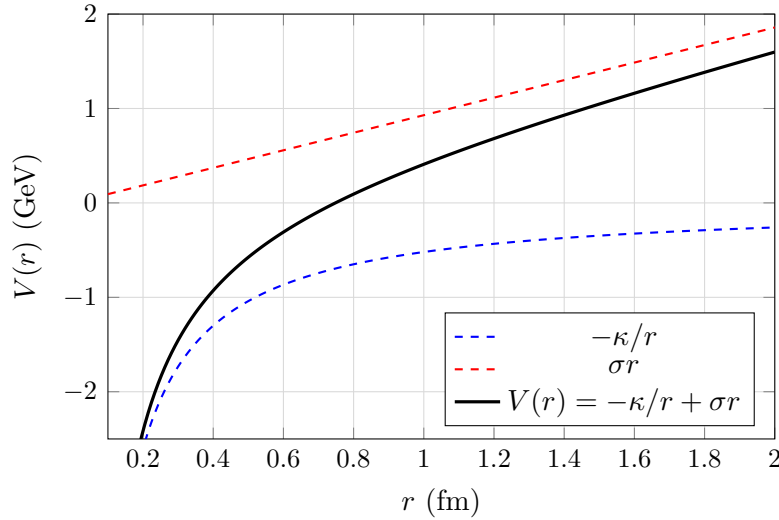


Figure 1.3: The Cornell potential as a function of interquark separation r . The dashed curves show the Coulomb ($-\kappa/r$, blue) and linear confinement (σr , red) components; the solid curve is their sum. At short distances the potential is dominated by one-gluon exchange; at large distances the linearly rising string tension ensures quark confinement.

1.3.3 Spin-dependent interactions

The spin-independent Cornell potential alone does not account for the full charmonium spectrum. The mass splitting between states with different spin quantum numbers—for instance, between the η_c (1S_0 , spin singlet) and the J/ψ (3S_1 , spin triplet)—requires spin-dependent corrections. At leading order, the dominant contribution for S -wave states is the hyperfine contact interaction [10],

$$V_{\text{hyp}}(r) = \frac{32\pi \alpha_s}{9 m_c^2} \delta^{(3)}(\mathbf{r}) \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \quad (1.20)$$

where α_s is the strong coupling constant and $\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$ is the spin–spin dot product. The eigenvalues of the latter operator distinguish singlet and triplet configurations:

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S = 0), \\ +\frac{1}{4} & \text{spin triplet } (S = 1). \end{cases} \quad (1.21)$$

Quarkonium states are classified in spectroscopic notation $^{2S+1}L_J$, where S is the total spin, L the orbital angular momentum ($S, P, D, \dots$ for $L = 0, 1, 2, \dots$), and J the total angular momentum. The low-lying charmonium states relevant to this work are:

- 1S_0 : spin singlet, S -wave $\rightarrow \eta_c$ meson ($M \approx 2984$ MeV),
- 3S_1 : spin triplet, S -wave $\rightarrow J/\psi$ meson ($M \approx 3097$ MeV),
- 1P_1 : spin singlet, P -wave $\rightarrow h_c$ meson ($M \approx 3526$ MeV).

1.3.4 Quarkonium as a quantum simulation testbed

The full quarkonium Hamiltonian, including kinetic energy and both spin-independent and spin-dependent potentials, reads

$$H = -\frac{1}{2\mu}\nabla^2 + V(r) + V_{\text{hyp}}(r), \quad (1.22)$$

where $\mu = m_c/2$ is the reduced mass. Expanding the radial wavefunction in a truncated basis of N quantum harmonic oscillator (QHO) eigenstates yields a finite-dimensional matrix eigenvalue problem of size $N \times N$, whose entries are determined by the matrix elements of the kinetic and potential operators in the QHO basis.

This matrix can then be decomposed into a sum of Pauli operators (Eq. (1.17)) and encoded on a quantum register of $\lceil \log_2 N \rceil$ qubits, making it directly amenable to variational quantum algorithms. With $N = 3$ – 4 basis states, the problem requires only 2–3 qubits, placing it firmly within reach of current NISQ devices while retaining the essential physics: multiple spectroscopic channels, spin-dependent splittings, and a hierarchy of excited states.

This combination of physical richness and computational tractability makes quarkonium an ideal testbed for benchmarking quantum simulation algorithms against exact solutions and experimental data.

1.4 Thesis Overview

This thesis investigates the application of variational quantum algorithms to the charmonium spectroscopy problem described above. We implement and compare several al-

algorithmic approaches for determining both ground and excited state energies across the 1S_0 , 3S_1 , and 1P_1 channels, assess their performance under realistic noise conditions, and evaluate error mitigation strategies. The goal is twofold: to demonstrate the feasibility of quantum simulation for hadronic physics on near-term hardware, and to provide a systematic comparison of methods that may inform future applications to more complex systems.

The thesis is organized as follows:

- **Chapter 2** develops the theoretical framework, including the Hamiltonian construction in the harmonic oscillator basis and the mapping to qubit operators.
- **Chapter 3** presents the quantum algorithms employed, covering both eigenvalue-based and time-evolution-based variational approaches.
- **Chapter 4** details the numerical results: energy spectra, convergence properties, and cross-method comparisons.
- **Chapter 5** addresses the effects of hardware noise and discusses error mitigation techniques.
- **Chapter 6** concludes with a summary of findings and an outlook on future directions.

Chapter 2

Mathematical Framework and Quantum Algorithms

This chapter develops the mathematical machinery that the remainder of the thesis relies on. We begin in Section 2.1 by constructing the charmonium Hamiltonian in a truncated harmonic-oscillator basis, treating the three channels of interest— 1S_0 , 3S_1 , and 1P_1 —on equal footing. Section 2.2 then encodes the resulting matrix on a quantum register through two complementary schemes: a compact direct mapping and a Jordan–Wigner mapping of an underlying second-quantised representation. The three quantum algorithms used throughout this work are introduced in turn: the Variational Quantum Eigensolver (Section 2.3), Variational Imaginary-Time Evolution (Section 2.4), and the probabilistic Trotter–Taylor Imaginary-Time Evolution scheme (Section 2.5). Beyond energy spectra, observable physical predictions also require transition amplitudes; Section 2.6 develops the quantum-circuit techniques for estimating M1 and E1 transitions between channels. We close (Section 2.7) with a side-by-side comparison of the three algorithms.

We follow the treatments of Gallimore and Liao [11], Woloshyn [12], and Yi et al. [13], adapting each to a common setting: the spin-dependent charmonium spectrum represented in a four-state harmonic-oscillator basis.

2.1 The Charmonium Hamiltonian

We begin by writing the charmonium Hamiltonian in a form amenable to numerical diagonalisation. The starting point is the non-relativistic quark model with a Cornell potential and a leading spin-dependent correction; the strategy is to expand the radial wavefunction in a finite harmonic-oscillator basis, producing a finite matrix whose lowest eigenvalues approximate the physical energy levels.

2.1.1 The non-relativistic quark model

A charmonium meson is a bound state of a charm quark and its antiquark [9, 10]. Because the charm mass $m_c \simeq 1.48 \text{ GeV}/c^2$ is large compared with the binding energy, the internal motion is well described non-relativistically: the relative coordinate $\mathbf{r} = \mathbf{r}_c - \mathbf{r}_{\bar{c}}$ satisfies the two-body Schrödinger equation

$$H \psi(\mathbf{r}) = \left[-\frac{\nabla^2}{2\mu} + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad (2.1)$$

with reduced mass $\mu = m_c/2$. The interaction $V(r)$ has two pieces. The first is the spin-independent Cornell potential

$$V_0(r) = -\frac{a}{r} + b r, \quad a = \frac{4\alpha_s}{3}, \quad (2.2)$$

where $-a/r$ is the colour-Coulomb attraction from one-gluon exchange and $b r$ is the linearly rising flux-tube term that confines the quarks [9]. The second piece is the leading spin-dependent correction. From the non-relativistic reduction of the Breit interaction, the dominant term at one-gluon-exchange order is the contact (hyperfine) interaction [10, 14]

$$V_s(r) = \frac{32\pi \alpha_s}{9 m_c^2} \delta_\sigma(\mathbf{r}) \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \quad (2.3)$$

where we have replaced the singular Dirac delta with a normalised Gaussian of width σ ,

$$\delta_\sigma(\mathbf{r}) = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 e^{-\sigma^2 r^2}, \quad (2.4)$$

to regularise the contact term inside our finite basis. The dot product of the two quark spin operators is fixed by the total spin $\mathbf{S} = \mathbf{S}_c + \mathbf{S}_{\bar{c}}$ through the identity $\mathbf{S}^2 = \mathbf{S}_c^2 + \mathbf{S}_{\bar{c}}^2 + 2 \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$, giving

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S = 0), \\ +\frac{1}{4} & \text{spin triplet } (S = 1). \end{cases} \quad (2.5)$$

The phenomenological parameters α_s , b , m_c , and σ used throughout this work are those of Barnes, Godfrey, and Swanson [14] and are listed in Table 2.1.

2.1.2 Radial equation and the three channels

Because $V_0(r)$ depends only on $r = |\mathbf{r}|$ and the spin operator in Eq. (2.3) commutes with the orbital angular momentum, eigenstates of H can be classified by the spectroscopic quantum numbers $^{2S+1}L_J$. Writing the wavefunction as

$$\psi(\mathbf{r}) = R_{n\ell}(r) Y_{\ell m}(\hat{\mathbf{r}}) \chi_{SM_S}, \quad (2.6)$$

Table 2.1: Cornell-potential parameters used in this work, following the charmonium fit of Barnes, Godfrey, and Swanson [14]. The conversion $\hbar c = 197.326$ MeV·fm relates inverse-femtometres to MeV. The Coulomb coefficient is $a = 4\alpha_s/3$.

Parameter	Symbol	Value
Strong coupling	α_s	0.5461
String tension	b	0.1425 GeV ²
Charm-quark mass	m_c	1.4794 GeV
Gaussian width	σ	1.0946 GeV

and defining the reduced radial function $u_{n\ell}(r) = rR_{n\ell}(r)$, the Schrödinger equation reduces to the one-dimensional eigenvalue problem

$$\left[-\frac{1}{2\mu} \frac{d^2}{dr^2} + V_0(r) + \frac{\ell(\ell+1)}{2\mu r^2} + \langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle \frac{32\pi\alpha_s}{9m_c^2} \delta_\sigma(r) \right] u_{n\ell}(r) = E_{n\ell} u_{n\ell}(r), \quad (2.7)$$

where the centrifugal barrier $\ell(\ell+1)/(2\mu r^2)$ is the standard remnant of the angular Laplacian, and the spin expectation value takes the discrete values of Eq. (2.5).

In this work we focus on the three low-lying channels that have served as benchmarks for quantum simulations of quarkonium [12]:

- 1S_0 (associated with the η_c meson): $\ell = 0$, $S = 0$. The centrifugal barrier vanishes and the contact term contributes with $\langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle = -3/4$.
- 3S_1 (associated with the J/ψ meson): $\ell = 0$, $S = 1$. As for 1S_0 but with $\langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle = +1/4$, producing the celebrated η_c - J/ψ hyperfine splitting.
- 1P_1 (associated with the h_c meson): $\ell = 1$, $S = 0$. The contact term contributes only at the origin, and because the $\ell = 1$ wavefunction vanishes there the spin-dependent term is numerically negligible for P -waves; the centrifugal barrier, however, shifts the energies up substantially.

Each channel thus generates a different one-dimensional radial Hamiltonian, which we now discretise.

2.1.3 Harmonic-oscillator basis expansion

The radial functions $u_{n\ell}(r)$ admit a convenient expansion in the eigenstates of a three-dimensional isotropic harmonic oscillator (HO),

$$\phi_{n\ell}(r) = N_{n\ell} r^{\ell+1} \exp\left(-\frac{r^2}{2b^2}\right) L_n^{\ell+1/2}\left(\frac{r^2}{b^2}\right), \quad (2.8)$$

with normalisation $N_{n\ell} = \sqrt{2n! b^{-(2\ell+3)} / \Gamma(n + \ell + 3/2)}$ and L_n^α the associated Laguerre polynomials. The length scale $b = (\mu\omega)^{-1/2}$ is determined by the oscillator frequency ω ,

which is chosen variationally so that the ground state of the basis already captures most of the physical ground-state wavefunction [11].

We truncate the basis to the first $N = 4$ states for each channel,

$$u_{n\ell}(r) \approx \sum_{k=0}^{N-1} c_k^{(n)} \phi_{k\ell}(r), \quad N = 4, \quad (2.9)$$

which turns Eq. (2.7) into the finite-dimensional eigenvalue problem

$$\sum_{k=0}^{N-1} H_{mk}^{(\ell,S)} c_k^{(n)} = E_n c_m^{(n)}, \quad (2.10)$$

where the channel-dependent matrix elements are

$$H_{mk}^{(\ell,S)} = \langle \phi_{m\ell} | T | \phi_{k\ell} \rangle + \langle \phi_{m\ell} | V_0 | \phi_{k\ell} \rangle + \langle \mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} \rangle \frac{32\pi\alpha_s}{9m_c^2} \langle \phi_{m\ell} | \delta_\sigma | \phi_{k\ell} \rangle. \quad (2.11)$$

Here T denotes the kinetic term plus, for $\ell > 0$, the centrifugal barrier.

Remark (Hylleraas–Undheim–MacDonald theorem). The truncation in Eq. (2.9) introduces a controlled approximation: the eigenvalues of the truncated matrix $H^{(\ell,S)} \in \mathbb{R}^{N \times N}$ are guaranteed to be upper bounds on the corresponding exact eigenvalues, and they are monotonically non-increasing in N [15]. Increasing N therefore always improves the variational estimate; $N = 4$ provides a useful balance between physical accuracy and the qubit cost of the encodings introduced in Section 2.2.

2.1.4 Matrix elements

The matrix elements in Eq. (2.11) can be evaluated in closed form using standard identities for the Laguerre polynomials and the hypergeometric function. We collect the key results here; complete derivations follow the methodology of Gallimore and Liao [11].

Kinetic energy. The kinetic operator, expressed through the ladder operators of the oscillator, is tridiagonal in the HO basis. For $\ell = 0$ one finds

$$\langle m | T | k \rangle = \frac{\omega}{2} \left[\left(2k + \frac{3}{2} \right) \delta_{mk} - \sqrt{k(k + \frac{1}{2})} \delta_{m,k-1} - \sqrt{(k+1)(k + \frac{3}{2})} \delta_{m,k+1} \right]. \quad (2.12)$$

For $\ell > 0$ the centrifugal term is absorbed into a shifted diagonal, $2k + \ell + 3/2$, with the off-diagonal couplings inheriting the same square-root structure with the index $\ell + 1/2$ in place of $1/2$.

Coulomb term. The matrix element of $1/r$ can be expressed in terms of a generalised hypergeometric function ${}_3F_2$ [11]:

$$\langle m|r^{-1}|k\rangle = \frac{(-1)^{m+k} 4b^{-1}}{\pi(1+2k)} \sqrt{\frac{\Gamma(m+\frac{3}{2})\Gamma(k+\frac{3}{2})}{m!k!}} {}_3F_2\left(\frac{1}{2}, 1, -m; \frac{3}{2}, \frac{1}{2}-k; 1\right). \quad (2.13)$$

Linear-confinement term. The matrix element of r involves the Gauss hypergeometric ${}_2F_1$:

$$\langle m|r|k\rangle = \frac{(-1)^{m+k} 4b}{\pi(1-4k^2)} \sqrt{\frac{\Gamma(m+\frac{3}{2})\Gamma(k+\frac{3}{2})}{m!k!}} {}_2F_1\left(2, -m; \frac{3}{2}-k; 1\right). \quad (2.14)$$

Contact term. For the regularised contact interaction with the Gaussian of Eq. (2.4), the matrix element reduces to a single Gaussian integral over a polynomial in r^2 , which is again closed-form:

$$\langle \phi_{m\ell} | \delta_\sigma | \phi_{k\ell} \rangle = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 \int_0^\infty \phi_{m\ell}(r) e^{-\sigma^2 r^2} \phi_{k\ell}(r) dr. \quad (2.15)$$

For $\ell > 0$ the factor $r^{2\ell+2}$ in the integrand suppresses the contribution near the origin; in the limit $\sigma \rightarrow \infty$ the integral vanishes for $\ell > 0$ and reproduces $|\phi_{m\ell}(0)\phi_{k\ell}(0)|$ for $\ell = 0$, recovering the unregularised Dirac-delta result.

Remark. The signs $(-1)^{m+k}$ in Eqs. (2.13) and (2.14) encode the alternating parity of the Laguerre polynomials at the origin and are essential for obtaining the correct off-diagonal Cornell-potential matrix elements. In practice all of these closed-form expressions are evaluated to high precision via the `mpmath` library and verified against direct numerical quadrature.

2.1.5 The three Hamiltonian matrices

Assembling Eq. (2.11) channel by channel with the parameters of Table 2.1 and $\omega = 1.2 \text{ fm}^{-1}$ produces the following three 4×4 matrices (in units of fm^{-1} , with conversion to

MeV via $\hbar c = 197.326 \text{ MeV} \cdot \text{fm}$):

$$H^{(1S_0)} = \begin{pmatrix} 0.9431 & -0.8733 & -0.7690 & -0.5601 \\ -0.8733 & 3.3365 & -0.5646 & -0.8648 \\ -0.7690 & -0.5646 & 5.4382 & -0.1566 \\ -0.5601 & -0.8648 & -0.1566 & 7.3451 \end{pmatrix}, \quad (2.16)$$

$$H^{(3S_1)} = \begin{pmatrix} 1.0946 & -0.7114 & -0.6111 & -0.4112 \\ -0.7114 & 3.5406 & -0.3910 & -0.6989 \\ -0.6111 & -0.3910 & 5.6122 & -0.0119 \\ -0.4112 & -0.6989 & -0.0119 & 7.5104 \end{pmatrix}, \quad (2.17)$$

$$H^{(1P_1)} = \begin{pmatrix} 2.8561 & -0.2395 & -0.3827 & -0.2282 \\ -0.2395 & 4.9190 & -0.0373 & -0.5097 \\ -0.3827 & -0.0373 & 6.8114 & 0.4058 \\ -0.2282 & -0.5097 & 0.4058 & 8.6034 \end{pmatrix}. \quad (2.18)$$

Each matrix is real and symmetric, a consequence of the Hermiticity of the underlying operator and the choice of real-valued HO basis functions. The 1P_1 matrix lies higher in absolute energy because of the $\ell = 1$ centrifugal contribution to the diagonal; the spin-dependent shift between 1S_0 and 3S_1 is visible already in the difference between their diagonal entries.

Remark (Errata in the literature). We note that in the published matrices for 1P_1 [12] the (3, 3) element was reported as 7.5104, coinciding numerically with the (3, 3) entry of $H^{(3S_1)}$. Cross-checking against the variational eigenvalues and the diagonal pattern of the kinetic + centrifugal + confinement contributions identifies this as a copy-paste error; the correct value 8.6034 is used in Eq. (2.18).

Diagonalising each $H^{(\ell,S)}$ by classical means immediately yields the exact (within the truncated basis) energy spectrum of its channel. Doing so on a quantum computer, however, requires representing the operator as a sum of Pauli strings acting on a register of qubits, which is the topic of the next section.

2.2 Encoding the Hamiltonian on Qubits

To run a quantum algorithm we must express the channel Hamiltonians Eqs. (2.16) to (2.18) in a form that the device can act on natively: a sum of tensor products of Pauli operators acting on a finite register of qubits. Two qualitatively different encoding choices appear in the literature for this kind of basis-expansion problem, and we will use both later in the thesis. This section derives them in parallel for the same underlying 4×4 matrices, allowing a direct comparison of qubit count and Pauli structure.

2.2.1 Pauli decomposition of a Hermitian matrix

Any Hermitian operator H on n qubits can be written as a real linear combination of the 4^n Pauli strings $\{P_k\} \subset \{I, X, Y, Z\}^{\otimes n}$,

$$H = \sum_{k=1}^{4^n} c_k P_k, \quad c_k = \frac{1}{2^n} \text{Tr}(H P_k) \in \mathbb{R}. \quad (2.19)$$

The relation Eq. (2.19) follows from the orthogonality of Pauli strings under the Hilbert–Schmidt inner product, $\text{Tr}(P_j P_k) = 2^n \delta_{jk}$. In practice only a small subset of the 4^n coefficients are nonzero; for real symmetric Hamiltonians, all strings containing an odd number of Y factors vanish identically, leaving 3^n candidate terms.

Once H has been put in the form Eq. (2.19), each Pauli string can be implemented as a basis rotation on the quantum register followed by a diagonal measurement, and the expectation value $\langle H \rangle$ recovered as the weighted sum $\sum_k c_k \langle P_k \rangle$. The remaining question is which qubit register to use—i.e., how to associate physical basis states with bit-strings. We discuss two natural choices.

2.2.2 Direct (compact) encoding

The most economical scheme assigns one bit-string to each HO basis function via the binary expansion of its index,

$$|n\rangle_{\text{HO}} \mapsto |\text{bin}(n)\rangle_{\text{qubit}}, \quad n = 0, 1, 2, 3. \quad (2.20)$$

For $N = 4$ this requires only $\lceil \log_2 N \rceil = 2$ qubits, and every computational-basis state of the register corresponds to a physical HO state. In particular,

$$|0\rangle \mapsto |00\rangle, \quad |1\rangle \mapsto |01\rangle, \quad |2\rangle \mapsto |10\rangle, \quad |3\rangle \mapsto |11\rangle. \quad (2.21)$$

The channel Hamiltonian $H^{(\ell, S)}$ is then literally a 4×4 matrix acting on this register, and its Pauli decomposition follows from Eq. (2.19). Applying that formula to the 1S_0 matrix of Eq. (2.16) yields, for example,

$$H^{(^1S_0)} = c_{II} II + c_{IZ} IZ + c_{ZI} ZI + c_{ZZ} ZZ + c_{IX} IX + c_{XI} XI + c_{ZX} ZX + c_{XZ} XZ + c_{XX} XX + c_{YY} YY, \quad (2.22)$$

i.e. ten Pauli terms, consistent with the absence of strings containing a single Y for a real-symmetric operator. The numerical coefficients $\{c_k\}$ are obtained by trace formulas and are real.

Example (Identity coefficient). Setting $P_k = I \otimes I$ in Eq. (2.19) gives the energy offset

$$c_{II} = \frac{1}{4} \text{Tr } H^{(1S_0)} = \frac{1}{4}(0.9431 + 3.3365 + 5.4382 + 7.3451) \approx 4.266,$$

which simply records the trace of the channel matrix and contributes only a constant shift to the eigenvalues.

The direct encoding has two clear advantages: it uses the minimal number of qubits, and every computational-basis state is physical, so the variational ansatz need not respect any external constraint. Its main drawback is that the resulting Pauli strings carry no obvious physical interpretation—they are simply the coefficients of an abstract 4×4 matrix.

2.2.3 Jordan–Wigner (occupation) encoding

The second encoding originates in quantum chemistry. We treat the HO basis functions as orbitals occupied by a single charm–anticharm pair and introduce a fermionic creation operator $a_n^\dagger$ for each orbital, with the canonical anticommutation relations

$$\{a_m, a_n^\dagger\} = \delta_{mn}, \quad \{a_m, a_n\} = \{a_m^\dagger, a_n^\dagger\} = 0. \quad (2.23)$$

The channel Hamiltonian is then the second-quantised one-body operator

$$\widehat{H}^{(\ell, S)} = \sum_{m, n=0}^{N-1} H_{mn}^{(\ell, S)} a_m^\dagger a_n, \quad (2.24)$$

acting on the Fock space of $N = 4$ orbitals; physical states are those containing exactly one excitation, i.e. the four states $\{a_n^\dagger |\text{vac}\rangle\}_{n=0}^3$. Restricting to this single-particle sector reproduces the original 4×4 matrix problem.

Jordan–Wigner transformation. The Jordan–Wigner (JW) transformation [2] maps each orbital to a dedicated qubit and faithfully reproduces the fermionic algebra Eq. (2.23) in terms of Pauli operators,

$$a_n^\dagger = \frac{1}{2} \left(\prod_{j < n} Z_j \right) (X_n - i Y_n), \quad (2.25)$$

$$a_n = \frac{1}{2} \left(\prod_{j < n} Z_j \right) (X_n + i Y_n). \quad (2.26)$$

The trailing string of Z operators on lower-indexed qubits encodes the fermionic sign that arises when a creation operator anticommutes past already-occupied orbitals. With this mapping, $N = 4$ orbitals require $N = 4$ qubits, of which only $\binom{4}{1} = 4$ computational-basis states are physical (namely $|0001\rangle, |0010\rangle, |0100\rangle, |1000\rangle$).

Pauli forms of the one-body operators. Substituting Eqs. (2.25) and (2.26) into the bilinears $a_m^\dagger a_n$ yields concrete Pauli strings. The number operator at orbital n is diagonal,

$$a_n^\dagger a_n = \frac{1}{2}(I - Z_n). \quad (2.27)$$

Adjacent hopping ($|m - n| = 1$, e.g. between orbitals n and $n + 1$) gives the familiar $XX+YY$ pattern,

$$a_n^\dagger a_{n+1} + a_{n+1}^\dagger a_n = \frac{1}{2}(X_n X_{n+1} + Y_n Y_{n+1}), \quad (2.28)$$

while non-adjacent hops pick up the JW Z -string,

$$a_n^\dagger a_{n+2} + a_{n+2}^\dagger a_n = \frac{1}{2}(X_n Z_{n+1} X_{n+2} + Y_n Z_{n+1} Y_{n+2}), \quad (2.29)$$

$$a_n^\dagger a_{n+3} + a_{n+3}^\dagger a_n = \frac{1}{2}(X_n Z_{n+1} Z_{n+2} X_{n+3} + Y_n Z_{n+1} Z_{n+2} Y_{n+3}). \quad (2.30)$$

Combining these expressions with the coefficients $H_{mn}^{(\ell,S)}$ from Eqs. (2.16) to (2.18) produces a Pauli-string decomposition of $\widehat{H}^{(\ell,S)}$.

Term counting. Collecting all distinct strings for $N = 4$, the Jordan–Wigner Hamiltonian contains

- one identity term (energy offset, from the diagonals through $a_n^\dagger a_n$);
- four single-qubit Z terms (one per orbital, from the diagonals);
- six pairs of $XX+YY$, $XZX+YZY$, or $XZZX+YZZY$ strings (one pair per off-diagonal element of $H^{(\ell,S)}$).

The total is $1 + 4 + 12 = 17$ Pauli strings, compared with the ten of the direct encoding. The two extra orbitals—and the corresponding Z -string overhead in Eq. (2.30)—are the price paid for the physically transparent occupation-number picture.

Remark (Reduction to Gallimore’s $N = 3$ case). Gallimore and Liao [11] applied the JW construction to a *spin-independent* Cornell potential truncated to $N = 3$ basis states, producing a 3×3 Hamiltonian on 3 qubits with 10 Pauli terms. The construction above extends that procedure in two directions: we include the spin-dependent contact term of Eq. (2.3) and we use $N = 4$ basis functions. Restricting to $N = 3$ and $\langle V_s \rangle = 0$ reproduces their decomposition exactly.

2.2.4 Comparison of the two encodings

Table 2.2 summarises the trade-offs. The direct encoding is more compact in qubit count and avoids any subspace constraint on the variational ansatz, whereas the JW encoding offers a transparent connection to second-quantised many-body methods and a natural

Table 2.2: Comparison of the two encodings of the channel Hamiltonians of Eqs. (2.16) to (2.18) on a quantum register of qubits.

	Direct encoding	Jordan–Wigner encoding
Qubits	$\lceil \log_2 N \rceil = 2$	$N = 4$
Physical states	all $2^2 = 4$	only $\binom{4}{1} = 4$ out of 16
Pauli terms	~ 10	~ 17
Ansatz constraint	none	single-particle sector
Physical picture	abstract	occupation of orbitals

framework for symmetry preservation. Both encodings represent exactly the same physical Hamiltonian in a different basis; the eigenvalues of Eqs. (2.16) to (2.18) are therefore identical in either picture. In what follows we adopt the direct encoding as the default for compactness, switching to the JW encoding when explicitly comparing it with Gallimore and Liao [11].

2.3 The Variational Quantum Eigensolver

With the Hamiltonian encoded as a Pauli sum, we now describe the simplest of the three algorithms used in this thesis: the Variational Quantum Eigensolver (VQE) [5, 6]. VQE is a direct application of the Rayleigh–Ritz principle in which the trial state is prepared by a parameterised quantum circuit and the energy is minimised by a classical optimiser. The role of the quantum computer is to evaluate the expectation value of an exponentially large Hamiltonian, a task that is in general intractable classically; the optimisation itself is performed in the much smaller classical parameter space.

2.3.1 The variational principle, in operator form

For any Hermitian Hamiltonian H on a finite Hilbert space, every normalised state $|\psi\rangle$ obeys

$$E_0 \leq \langle \psi | H | \psi \rangle, \quad (2.31)$$

where E_0 is the smallest eigenvalue of H , with equality if and only if $|\psi\rangle$ lies in the corresponding eigenspace. Restricting the search to a family of *trial states* $\{|\psi(\boldsymbol{\theta})\rangle\}$ parameterised by a classical vector $\boldsymbol{\theta} \in \mathbb{R}^p$ gives the upper bound

$$E_0 \leq \min_{\boldsymbol{\theta}} E(\boldsymbol{\theta}), \quad E(\boldsymbol{\theta}) \equiv \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle. \quad (2.32)$$

The VQE algorithm computes $E(\boldsymbol{\theta})$ on the quantum processor and minimises it on a classical computer; the resulting parameters $\boldsymbol{\theta}^*$ characterise an approximation $|\psi(\boldsymbol{\theta}^*)\rangle$ to the ground state.

The accuracy of this approximation is controlled by the *expressibility* of the ansatz: if the parameterised manifold $\{|\psi(\boldsymbol{\theta})\rangle\}$ contains the true ground state then the minimum of Eq. (2.32) coincides with E_0 ; otherwise it provides a (variational) upper bound. Two ansatz constructions are particularly natural for the encodings of Section 2.2.

2.3.2 Ansatz for the direct encoding

For the two-qubit direct encoding of Section 2.2.2, a state of the form

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle, \quad c_n \in \mathbb{R}, \quad \sum_n c_n^2 = 1, \quad (2.33)$$

has three independent real degrees of freedom (four amplitudes minus the normalisation). Because the channel Hamiltonians of Eqs. (2.16) to (2.18) are real and symmetric, their eigenstates can be chosen real, and a three-parameter circuit producing the most general real state of the form Eq. (2.33) is therefore expressible enough. Such a circuit, due to Wołoszyn [12], applies two single-qubit R_y rotations, a CNOT, and a third R_y :

$$U(\boldsymbol{\theta}) = [I \otimes R_y(\theta_2)] \text{CNOT}_{0 \rightarrow 1} [R_y(\theta_1) \otimes R_y(\theta_0)], \quad (2.34)$$

with $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)$. The corresponding circuit is shown in Fig. 2.1.

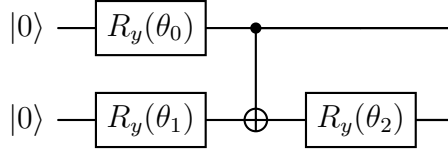


Figure 2.1: Three-parameter ansatz used with the direct encoding. The two initial R_y rotations create an arbitrary product state, the CNOT entangles the qubits, and the final R_y provides the third independent degree of freedom required to reach a general real four-amplitude state.

Acting on $|00\rangle$, this circuit yields amplitudes

$$|\psi(\boldsymbol{\theta})\rangle = \begin{pmatrix} c_0(c_1c_2 - s_1s_2) \\ s_0(s_1c_2 - c_1s_2) \\ c_0(c_1s_2 + s_1c_2) \\ s_0(s_1s_2 + c_1c_2) \end{pmatrix}, \quad c_n = \cos(\theta_n/2), \quad s_n = \sin(\theta_n/2), \quad (2.35)$$

which can be verified to cover the unit sphere in $\mathbb{R}^4$ as $\boldsymbol{\theta}$ ranges over a domain of size $4\pi^3$, modulo redundancies.

2.3.3 Ansatz for the Jordan–Wigner encoding

For the four-qubit Jordan–Wigner encoding of Section 2.2.3 the trial state must remain within the single-particle sector. A natural construction starts from the reference state $|0001\rangle$ (one excitation in the lowest orbital) and applies a unitary built from particle-conserving operators. Extending the Unitary Coupled Cluster (UCC) ansatz of Gallimore and Liao [11] from $N = 3$ to $N = 4$ basis states, we define

$$U_{\text{UCC}}(\theta_1, \theta_2, \theta_3) = \exp \left\{ \sum_{k=1}^3 \theta_k (a_k^\dagger a_0 - a_0^\dagger a_k) \right\}. \quad (2.36)$$

Each generator $a_k^\dagger a_0 - a_0^\dagger a_k$ is anti-Hermitian and preserves particle number, so U_{UCC} is unitary and maps single-particle states to single-particle states. With three independent parameters the operator generates the most general real superposition in the single-particle sector,

$$|\psi(\alpha, \beta, \gamma)\rangle = \cos \alpha |0001\rangle + \sin \alpha \cos \beta |0010\rangle + \sin \alpha \sin \beta \cos \gamma |0100\rangle + \sin \alpha \sin \beta \sin \gamma |1000\rangle, \quad (2.37)$$

which is simply the $\mathbb{S}^3$ spherical parameterisation lifted into the single-particle sector. The trigonometric reparameterisation $(\theta_1, \theta_2, \theta_3) \mapsto (\alpha, \beta, \gamma)$ is closed-form but rarely needed in practice; the circuit is built directly from Eq. (2.36) using a sequence of Givens rotations between adjacent orbitals [11]. Compared with the direct-encoding ansatz of Fig. 2.1, the UCC ansatz uses four qubits and a deeper circuit, but it has the conceptual advantage of representing physical particle hopping between HO orbitals.

2.3.4 Measuring the energy on a quantum processor

Given an ansatz $|\psi(\boldsymbol{\theta})\rangle$ and the Pauli decomposition $H = \sum_k c_k P_k$ from Eq. (2.19), the energy estimator is linear in the Pauli expectation values,

$$\hat{E}(\boldsymbol{\theta}) = \sum_k c_k \langle P_k \rangle_{\boldsymbol{\theta}}, \quad (2.38)$$

where each $\langle P_k \rangle_{\boldsymbol{\theta}}$ is estimated separately by preparing $|\psi(\boldsymbol{\theta})\rangle$ many times and measuring P_k .

A measurement in the computational basis directly gives access to $\langle Z_i \rangle$, but X and Y observables require a basis rotation applied before the measurement. The rotations are obtained from the eigenbasis identities $X = HZH$ and $Y = (SH)Z(HS^\dagger)$, giving the

standard recipe

$$\langle X_i \rangle = \text{measure } Z_i \text{ after applying } H \text{ on qubit } i, \quad (2.39)$$

$$\langle Y_i \rangle = \text{measure } Z_i \text{ after applying } S^\dagger H \text{ on qubit } i. \quad (2.40)$$

Each Pauli string P_k is mapped to a diagonal observable by applying these single-qubit basis rotations to every non-identity factor before the final measurement.

Remark (Shot noise). With a finite number N_{shots} of repetitions per circuit, the estimator Eq. (2.38) fluctuates around its true value with a statistical uncertainty $\sigma_{\hat{E}} = \sqrt{\sum_k c_k^2 \sigma_{P_k}^2 / N_{\text{shots}}}$, where $\sigma_{P_k}^2 \leq 1$ for any Pauli. The relative error therefore decreases only as $1/\sqrt{N_{\text{shots}}}$, which makes shot count an important resource and motivates the choice of optimiser discussed next.

2.3.5 Classical optimisation

The cost function $\hat{E}(\boldsymbol{\theta})$ is, in general, non-convex and contaminated by shot noise. Two broad families of classical optimisers are available:

- **Gradient-free methods** (e.g. COBYLA, Nelder–Mead) only require evaluations of $\hat{E}$ and are robust to moderate statistical noise. They typically converge in tens to a few hundreds of iterations for low-dimensional parameter spaces.
- **Gradient-based methods**, which require partial derivatives of $\hat{E}$ at the cost of additional circuit evaluations. The *parameter-shift rule* (Section 2.4) provides exact analytic gradients for rotation gates, but each gradient component doubles the number of circuits.

For the few-parameter ansatzes of Sections 2.3.2 and 2.3.3 we use COBYLA throughout, which balances simplicity, noise tolerance, and convergence speed. Figure 2.2 recalls the resulting hybrid loop already introduced in Chapter 1: the classical optimiser proposes a parameter update, the quantum processor evaluates $\hat{E}(\boldsymbol{\theta})$, and the cycle repeats until convergence.

2.3.6 Excited states via orthogonalisation

VQE in the form above targets only the ground state. To reach excited levels, we constrain the trial state to be orthogonal to the previously obtained eigenstates. Suppose that the ground state $|\psi_0(\boldsymbol{\theta}_0^*)\rangle$ has been determined; the first excited state is then obtained by minimising

$$E_1 = \min_{\boldsymbol{\theta}: \langle \psi_0 | \psi(\boldsymbol{\theta}) \rangle = 0} \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle. \quad (2.41)$$

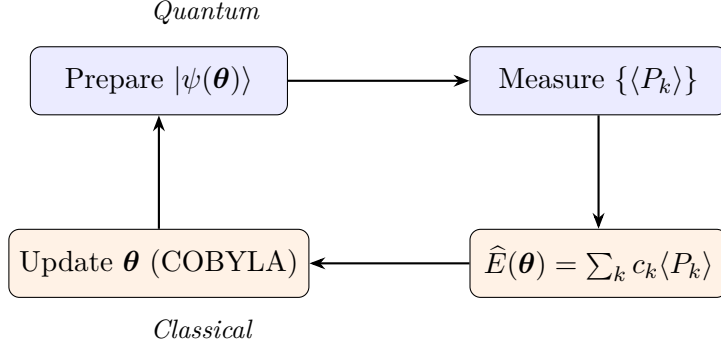


Figure 2.2: The VQE hybrid loop specialised to charmonium spectroscopy. The quantum processor prepares the ansatz of Fig. 2.1 or its UCC counterpart, measures the Pauli expectation values in Eq. (2.38), and returns the energy estimate to a classical optimiser.

For the two-qubit direct ansatz this constrained optimisation is one-dimensional: the orthogonal complement of $|\psi_0\rangle$ inside the ansatz manifold is parameterised by a single angle γ , and the optimisation reduces to a one-dimensional scan. Gallimore and Liao [11] provide an explicit parametrisation for $N = 3$ which generalises straightforwardly to $N = 4$: the orthogonal trial state takes the form

$$|\psi_1(\gamma)\rangle = \cos \gamma |\psi_0^{\perp,a}\rangle + \sin \gamma |\psi_0^{\perp,b}\rangle, \quad (2.42)$$

where $|\psi_0^{\perp,a}\rangle, |\psi_0^{\perp,b}\rangle$ are two fixed states orthogonal to $|\psi_0\rangle$ that span the rest of the real ansatz manifold. Iterating the procedure, each new excited state is constrained to be orthogonal to all previously obtained states.

Remark (Error propagation). A useful property of the orthogonalisation method is that errors in the ground-state estimate propagate in a bounded way: if the variational ground state has an error $\delta_0 = E_0^{\text{VQE}} - E_0$, the first excited state estimate E_1^{VQE} satisfies $|E_1^{\text{VQE}} - E_1| \leq \delta_0$ [11], so the accuracy does not deteriorate as we move up the spectrum.

The orthogonalisation approach exemplifies a recurring theme of variational methods: complex spectral information is obtained by repeatedly solving a sequence of constrained low-dimensional optimisation problems, each implemented as a small quantum circuit plus a classical optimiser. The next section presents a complementary strategy in which the same parameterised ansatz is evolved in imaginary time rather than directly optimised.

2.4 Variational Imaginary-Time Evolution

VQE locates the ground state by directly minimising the Rayleigh quotient. An alternative strategy is to simulate the (non-unitary) imaginary-time Schrödinger equation, whose long-time limit projects any initial state with non-vanishing ground-state overlap onto the ground state itself. Implementing this evolution exactly is incompatible with the unitar-

ity of quantum circuits; the variational imaginary-time evolution (VQITE) algorithm of McArdle et al. [16] and Yuan et al. [17] circumvents this by representing the evolved state with the same kind of parameterised circuit used in VQE and deriving classical equations of motion for the parameters from a variational principle. The result is a sequence of small linear systems—each constructed from quantities that the quantum processor can estimate—that drive the ansatz toward the ground state.

2.4.1 Imaginary-time evolution as a ground-state projector

The imaginary-time Schrödinger equation is obtained from its real-time counterpart by the Wick rotation $t \mapsto -i\tau$, with $\tau \geq 0$, giving

$$\frac{d|\psi(\tau)\rangle}{d\tau} = -H|\psi(\tau)\rangle. \quad (2.43)$$

Its (unnormalised) solution is $e^{-H\tau}|\psi(0)\rangle$. Expanding the initial state in the energy eigenbasis, $|\psi(0)\rangle = \sum_k a_k |E_k\rangle$, gives

$$e^{-H\tau}|\psi(0)\rangle = \sum_k a_k e^{-E_k\tau} |E_k\rangle. \quad (2.44)$$

Factoring out the ground-state contribution and renormalising,

$$|\psi(\tau)\rangle = \frac{a_0 |E_0\rangle + \sum_{k>0} a_k e^{-(E_k-E_0)\tau} |E_k\rangle}{\sqrt{|a_0|^2 + \sum_{k>0} |a_k|^2 e^{-2(E_k-E_0)\tau}}} \xrightarrow{\tau \rightarrow \infty} \frac{a_0}{|a_0|} |E_0\rangle, \quad (2.45)$$

provided $a_0 = \langle E_0 | \psi(0) \rangle \neq 0$. The excited-state amplitudes are exponentially suppressed at a rate set by the gap $E_k - E_0$, so the algorithm reaches the ground state in a number of imaginary-time steps proportional to $1/(E_1 - E_0)$.

To preserve the norm dynamically, it is convenient to subtract the instantaneous energy, $E_\tau \equiv \langle \psi(\tau) | H | \psi(\tau) \rangle$, from H . The *normalised* imaginary-time equation

$$\frac{d|\psi(\tau)\rangle}{d\tau} = -(H - E_\tau)|\psi(\tau)\rangle \quad (2.46)$$

preserves the norm of $|\psi\rangle$ and has the same long- τ limit as Eq. (2.43), but is more convenient to discretise variationally.

Remark (Non-unitarity). The operator $e^{-H\tau}$ is Hermitian and positive definite, but it is not unitary: $(e^{-H\tau})^\dagger e^{-H\tau} = e^{-2H\tau} \neq I$. Quantum circuits, by contrast, are unitary by construction. Implementing Eq. (2.46) on a quantum processor therefore requires an indirect strategy; VQITE projects the exact non-unitary dynamics onto a unitary parameterised submanifold, while Section 2.5 will encode it probabilistically using ancillas.

2.4.2 The McLachlan variational principle

VQITE replaces the exact evolution Eq. (2.46) by a *parameterised* trajectory $|\psi(\boldsymbol{\theta}(\tau))\rangle$ on the same ansatz manifold used by VQE. The parameters $\boldsymbol{\theta}(\tau)$ are evolved so that the parameterised state tracks the exact one as closely as possible. We adopt the principle of McLachlan [18], which minimises at each instant the distance between the actual time-derivative of the trial state and the right-hand side of Eq. (2.46):

$$\delta_{\dot{\boldsymbol{\theta}}} \left\| \frac{\partial}{\partial \tau} |\psi(\boldsymbol{\theta}(\tau))\rangle + (H - E_{\tau}) |\psi(\boldsymbol{\theta}(\tau))\rangle \right\|^2 = 0. \quad (2.47)$$

Here the variation is taken with respect to the time derivative $\dot{\boldsymbol{\theta}} \equiv d\boldsymbol{\theta}/d\tau$, treated as an independent vector. Expanding $\partial_{\tau} |\psi(\boldsymbol{\theta}(\tau))\rangle = \sum_i \dot{\theta}_i |\partial_i \psi\rangle$, where $|\partial_i \psi\rangle \equiv \partial |\psi(\boldsymbol{\theta})\rangle / \partial \theta_i$, Eq. (2.47) becomes a quadratic form in $\dot{\boldsymbol{\theta}}$. Carrying out the variation gives the linear system

$$\sum_j A_{ij} \dot{\theta}_j = C_i, \quad i = 1, \dots, p, \quad (2.48)$$

with matrix and vector elements

$$A_{ij} = \Re \left[\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle \right], \quad (2.49)$$

$$C_i = -\Re \left[\langle \partial_i \psi | H | \psi \rangle \right] + \Re \left[\langle \partial_i \psi | \psi \rangle \right] E_{\tau}. \quad (2.50)$$

For ansatz circuits that preserve the global phase of $|\psi\rangle$ (which is the case for the constructions of Sections 2.3.2 and 2.3.3) the second term in C_i vanishes and C_i reduces to a real-valued energy gradient,

$$C_i = -\frac{1}{2} \frac{\partial E(\boldsymbol{\theta})}{\partial \theta_i}. \quad (2.51)$$

The matrix A is precisely the (real) Fubini–Study quantum metric on the ansatz manifold; it is real, symmetric, and positive semidefinite, and it plays the same role for variational quantum evolution as the Fisher information matrix in classical statistics [19].

2.4.3 The quantum metric tensor in practice

The geometric interpretation of A_{ij} is illuminating, but more useful for implementation is its expression as a curvature of the *fidelity surface*

$$f(\boldsymbol{\theta}, \boldsymbol{\theta}') \equiv |\langle \psi(\boldsymbol{\theta}) | \psi(\boldsymbol{\theta}') \rangle|^2. \quad (2.52)$$

Expanding around $\boldsymbol{\theta}' = \boldsymbol{\theta}$ to second order yields

$$f(\boldsymbol{\theta}, \boldsymbol{\theta} + \boldsymbol{\delta}) = 1 - \sum_{ij} A_{ij} \delta_i \delta_j + O(\delta^3), \quad (2.53)$$

so that the metric coincides with the leading curvature of the fidelity landscape. This identification is the basis of the standard finite-difference estimator: from Eq. (2.53),

$$A_{ij} \approx -\frac{1}{4\varepsilon^2} \left[f(\boldsymbol{\theta} + \varepsilon e_i + \varepsilon e_j) - f(\boldsymbol{\theta} + \varepsilon e_i - \varepsilon e_j) - f(\boldsymbol{\theta} - \varepsilon e_i + \varepsilon e_j) + f(\boldsymbol{\theta} - \varepsilon e_i - \varepsilon e_j) \right], \quad (2.54)$$

with e_i the unit vector in the i -direction. Each of the four fidelities is the absolute square of an overlap between two ansatz states, which can be estimated on a quantum processor either by the swap test or by the $|\psi_2\rangle^\dagger |\psi_1\rangle$ inverse-ansatz protocol described in Section 2.6. For the $p = 3$ ansatze used here, the symmetric metric requires $p(p+1)/2 = 6$ independent entries, each costing four overlap measurements at a small step $\varepsilon \sim 10^{-3}$ – 10^{-4} .

2.4.4 The energy gradient and the parameter-shift rule

The vector C_i in Eq. (2.51) is a derivative of an expectation value with respect to a circuit parameter. For ansatze composed of single-qubit rotations of the form $R_\mu(\theta) = e^{-i\theta\mu/2}$ with $\mu \in \{X, Y, Z\}$, such derivatives admit a remarkable closed-form expression—the *parameter-shift rule* [20, 21]:

$$\frac{\partial \langle H \rangle_{\boldsymbol{\theta}}}{\partial \theta_i} = \frac{1}{2} \left[\langle H \rangle_{\boldsymbol{\theta} + (\pi/2) e_i} - \langle H \rangle_{\boldsymbol{\theta} - (\pi/2) e_i} \right]. \quad (2.55)$$

The remarkable feature of Eq. (2.55) is that it is *exact*—unlike a finite-difference derivative, the formula returns the true partial derivative regardless of the step size. The proof, which we sketch here, illustrates the special role played by Pauli rotations.

Remark (Sketch of the parameter-shift derivation). Consider a circuit where the parameter θ_i appears only inside a rotation $R_\mu(\theta_i) = e^{-i\theta_i\mu/2}$. Since $\mu^2 = I$, we have $R_\mu(\theta_i) = \cos(\theta_i/2) I - i \sin(\theta_i/2) \mu$, so the dependence of R_μ —and hence of $|\psi(\boldsymbol{\theta})\rangle$ —on θ_i is restricted to the linear span of $\{1, \cos \theta_i, \sin \theta_i\}$. Therefore the expectation value $\langle H \rangle_{\boldsymbol{\theta}}$ is a real-valued trigonometric polynomial of period 2π in θ_i , of the form $a \cos \theta_i + b \sin \theta_i + c$. Setting the right-hand side of Eq. (2.55) equal to $b \cos \theta_i - a \sin \theta_i$, which is exactly the derivative of $a \cos \theta_i + b \sin \theta_i + c$, completes the proof.

The practical implication is that, for ansatz circuits built from Pauli rotations, all partial derivatives required by VQITE—i.e. each component of C —can be evaluated on the quantum processor at a cost of two additional energy estimations per parameter. For the three-parameter ansatze used in this work, the cost is just six extra circuit evaluations per imaginary-time step.

2.4.5 Time stepping and regularisation

Combining Eqs. (2.48), (2.49) and (2.51) gives the equation of motion for $\boldsymbol{\theta}$:

$$\dot{\boldsymbol{\theta}} = A^{-1} C. \quad (2.56)$$

Discretising with the forward Euler scheme yields a simple update rule for the parameter vector after a small imaginary-time step $\Delta\tau$,

$$\boldsymbol{\theta}(\tau + \Delta\tau) = \boldsymbol{\theta}(\tau) + \Delta\tau A^{-1}(\boldsymbol{\theta}) C(\boldsymbol{\theta}). \quad (2.57)$$

Two practical issues arise. First, when two parameters become approximately degenerate (e.g., when the ansatz fails to be locally injective), the metric A becomes ill-conditioned. We add a small Tikhonov regulariser to restore numerical stability,

$$(A + \lambda I) \dot{\boldsymbol{\theta}} = C, \quad \lambda \sim 10^{-4}. \quad (2.58)$$

Second, the explicit Euler scheme is conditionally stable and accuracy is limited by the curvature of the energy landscape. Step sizes around $\Delta\tau \simeq 0.02$ in natural units of $1/\|H\|$ are sufficient to reach a stable ground state in 50–100 steps for the channel Hamiltonians of Eqs. (2.16) to (2.18).

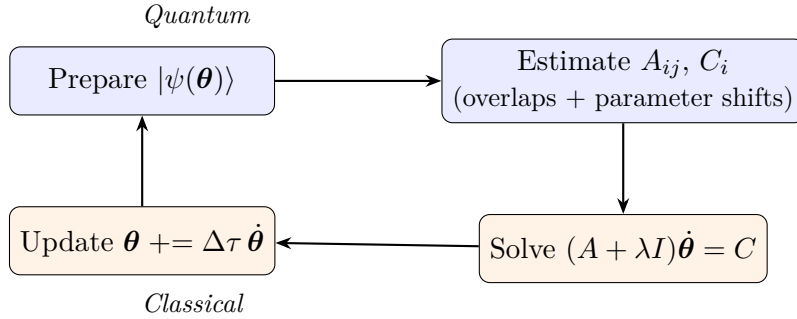


Figure 2.3: The VQITE hybrid loop. At each step the quantum processor estimates the metric tensor A via Eq. (2.54) and the energy gradient components of C via the parameter-shift rule Eq. (2.55); the classical computer solves the linear system Eq. (2.58) and advances the parameters in imaginary time.

2.4.6 Excited states by penalisation

VQITE projects onto the lowest-energy state with non-vanishing overlap with the initial state. To target higher levels we modify the effective Hamiltonian by adding penalty terms that suppress overlap with previously obtained states. Concretely, if $\{|\psi_k^*\rangle\}_{k=0}^{n-1}$ are

already-computed eigenstates, we replace H by

$$H_{\text{eff}} = H + \alpha \sum_{k=0}^{n-1} |\psi_k^*\rangle \langle \psi_k^*|, \quad (2.59)$$

where α is a positive penalty large enough to shift the energies of the already-found states above the next eigenvalue. Running VQITE with H_{eff} then converges to the next-higher eigenstate. The energy gradient generalises to

$$\frac{\partial \langle H_{\text{eff}} \rangle}{\partial \theta_i} = \frac{\partial \langle H \rangle}{\partial \theta_i} + 2\alpha \sum_k \Re[\langle \psi_k^* | \psi \rangle \langle \partial_i \psi | \psi_k^* \rangle], \quad (2.60)$$

where the overlap correction is again accessible from short fidelity circuits. In practice α is chosen of the same order of magnitude as the typical gap between adjacent eigenvalues; we use $\alpha = 10$ in inverse-femtometre units throughout this thesis.

The combination of VQITE for the ground state with iterated penalisation provides access to a sequence of eigenpairs without ever requiring the diagonalisation of an exponentially large matrix on a classical computer. The cost is a sequence of linear systems of size at most $p = 3$, an imaginary-time integration with $O(50)$ steps per state, and a number of small overlap circuits scaling polynomially with the number of states targeted.

2.5 Trotter–Taylor Imaginary-Time Evolution

The two methods discussed so far—VQE in Section 2.3 and VQITE in Section 2.4—share a common strategy: a parameterised ansatz approximates the target state, and a classical optimiser navigates the parameter space. This restricts the accuracy to the expressibility of the chosen ansatz. The Trotter–Taylor Imaginary-Time Evolution (TTITE) scheme of Yi et al. [13], by contrast, simulates the imaginary-time propagator $e^{-H\tau}$ directly on the quantum register, without ever fitting parameters. The price is a probabilistic structure: at each step an ancilla measurement must yield a particular outcome, and the algorithm proceeds only when it does. This section develops the construction in stages, applying it throughout to the channel Hamiltonians of Eqs. (2.16) to (2.18).

2.5.1 Strategy: implement the propagator directly

The starting point is again the eigenstate-projection property of imaginary time Eq. (2.45). We seek a unitary protocol that, given the initial state $|\psi(0)\rangle$ on the work register, returns $|\psi(\tau)\rangle \propto e^{-H\tau} |\psi(0)\rangle$ to controllable accuracy. Two obstacles must be overcome.

The first obstacle is the non-unitarity of the propagator $e^{-H\tau}$. We resolve it by writing each elementary factor of a Trotter-decomposed propagator as a linear combination of two unitaries; such “LCU” operators admit a probabilistic implementation via an ancilla.

The second obstacle is that the Hamiltonian is a *sum* of Pauli strings, and exponentials of a sum cannot generally be separated. We resolve it through the Trotter formula, which decomposes $e^{-H\Delta\tau}$ into a product of exponentials of individual Pauli strings, each of which is straightforward to handle.

We now make each of these steps precise.

2.5.2 Trotter decomposition

Let $H = \sum_{i=1}^m c_i h_i$ be the Pauli decomposition of one of the channel Hamiltonians, with $c_i \in \mathbb{R}$ and each h_i a Pauli string satisfying $h_i^2 = I$. The first-order *Trotter formula* [22, 23] approximates the imaginary-time propagator for a small step $\Delta\tau$ as

$$e^{-H\Delta\tau} = \prod_{i=1}^m e^{-c_i h_i \Delta\tau} + O((\Delta\tau)^2), \quad (2.61)$$

with leading error proportional to the commutators $[h_i, h_j]$. To evolve over a total imaginary time τ we apply $L = \tau/\Delta\tau$ such Trotter steps in succession, accumulating an overall error of order $\tau\Delta\tau$. Higher-order symmetric (Suzuki) schemes reduce this error to $O(\Delta\tau^{2k+1})$ for stage- $2k$ formulas, at the cost of more elementary exponentials per step [23]; for the small Hamiltonians of this work the first-order formula Eq. (2.61) is already accurate enough.

2.5.3 Taylor expansion of a single Trotter step

Each factor $e^{-c h \Delta\tau}$ in Eq. (2.61) is the exponential of an Hermitian operator whose square is the identity, $h^2 = I$. This property gives a remarkably simple closed form. Expanding the exponential as a power series and separating even and odd powers,

$$\begin{aligned} e^{-c h \Delta\tau} &= \sum_{j=0}^{\infty} \frac{(-c \Delta\tau)^j}{j!} h^j \\ &= \left(\sum_{j \text{ even}} \frac{(-c \Delta\tau)^j}{j!} \right) I + \left(\sum_{j \text{ odd}} \frac{(-c \Delta\tau)^j}{j!} \right) h = \cosh(c \Delta\tau) I - \sinh(c \Delta\tau) h, \end{aligned} \quad (2.62)$$

$$(2.63)$$

where we used $h^{2k} = I$ and $h^{2k+1} = h$. The single Trotter step is therefore exactly a linear combination of two unitaries: the identity and h itself. Truncating the series at finite order R ,

$$T^{(R)}(c \Delta\tau) \equiv \alpha I + \beta h, \quad \alpha = \sum_{\substack{j=0 \\ j \text{ even}}}^R \frac{(-c \Delta\tau)^j}{j!}, \quad \beta = \sum_{\substack{j=1 \\ j \text{ odd}}}^R \frac{(-c \Delta\tau)^j}{j!}, \quad (2.64)$$

incurs a Taylor error of order $(\Delta\tau)^{R+1}$, which for $R = 4$ or higher is negligible compared to the Trotter error $O(\Delta\tau^2)$ in Eq. (2.61). The truncation is convenient because it makes α and β polynomials in $\Delta\tau$, slightly simpler than $\cosh$ and $\sinh$.

Remark. The closed form Eq. (2.63) shows that the Taylor truncation is in fact *not necessary* for correctness; one could simply use $\alpha = \cosh(c \Delta\tau)$ and $\beta = -\sinh(c \Delta\tau)$. The finite-order form Eq. (2.64) matches the construction of Yi et al. [13] and makes the analysis of the LCU success probability slightly more uniform across different Hamiltonians, but the algorithm itself is unchanged.

2.5.4 Linear-combination-of-unitaries (LCU) implementation

The operator $\alpha I + \beta h$ in Eq. (2.64) is a non-unitary linear combination of two unitaries. Operators of this form admit a celebrated probabilistic implementation, the LCU construction [24], which uses a single ancillary qubit. We derive it explicitly here.

Augment the work register with one ancilla qubit, initially in $|0\rangle$. The combined state is $|0\rangle \otimes |\psi\rangle$. The LCU circuit applies the following three operations.

(i) Ancilla preparation. Apply $R_y(\vartheta)$ to the ancilla with

$$\vartheta = 2 \operatorname{atan2}(\beta, \alpha), \quad (2.65)$$

so that $R_y(\vartheta) |0\rangle = \cos(\vartheta/2) |0\rangle + \sin(\vartheta/2) |1\rangle = (\alpha |0\rangle + \beta |1\rangle) / \sqrt{\alpha^2 + \beta^2}$. The combined state is then

$$|\phi_1\rangle = \frac{1}{\sqrt{\alpha^2 + \beta^2}} (\alpha |0\rangle + \beta |1\rangle) \otimes |\psi\rangle. \quad (2.66)$$

(ii) Controlled application of h . Apply the controlled unitary $CU = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes h$, which acts as the identity if the ancilla is $|0\rangle$ and as h if it is $|1\rangle$. Since h is a Pauli string of length n , the controlled gate decomposes into n elementary controlled-Pauli gates acting on the work qubits. After this step,

$$|\phi_2\rangle = \frac{1}{\sqrt{\alpha^2 + \beta^2}} (\alpha |0\rangle \otimes |\psi\rangle + \beta |1\rangle \otimes h |\psi\rangle). \quad (2.67)$$

(iii) Hadamard and measurement on the ancilla. Apply a Hadamard gate to the ancilla and then measure it in the computational basis. Using $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$, the state just before measurement is

$$|\phi_3\rangle = \frac{1}{\sqrt{2(\alpha^2 + \beta^2)}} \left(|0\rangle \otimes (\alpha I + \beta h) |\psi\rangle + |1\rangle \otimes (\alpha I - \beta h) |\psi\rangle \right). \quad (2.68)$$

A measurement outcome of 0 on the ancilla projects the work register onto the (unnormalised) state $(\alpha I + \beta h) |\psi\rangle$. After renormalisation this is precisely the desired truncated

Trotter step.

The probability of measuring $|0\rangle$ on the ancilla,

$$P_s = \frac{\|(\alpha I + \beta h)|\psi\rangle\|^2}{2(\alpha^2 + \beta^2)}, \quad (2.69)$$

sets the cost of the protocol: if the measurement returns $|1\rangle$ the work state has been corrupted and the segment must be restarted from $|\psi\rangle$. For small $\Delta\tau$ one has $\alpha \approx 1$ and $\beta \approx -c\Delta\tau$, so $P_s \approx 1/2$; the protocol is therefore efficient at small step sizes.

Figure 2.4 summarises the LCU circuit graphically.

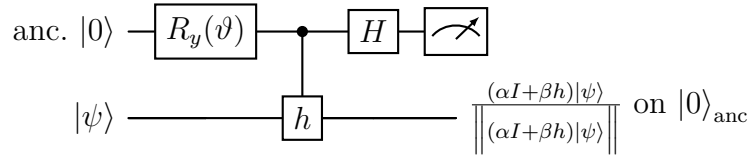


Figure 2.4: LCU circuit implementing one truncated Trotter step $T^{(R)}(c\Delta\tau) = \alpha I + \beta h$ on the work register $|\psi\rangle$. The ancilla is rotated by an angle $\vartheta = 2\text{atan2}(\beta, \alpha)$, a controlled application of the Pauli string h entangles ancilla and work, a Hadamard interferes the two LCU branches, and a measurement on the ancilla post-selects the desired branch. The work register collapses to the (renormalised) truncated step exactly when the measurement outcome is 0, an event of probability P_s given by Eq. (2.69).

2.5.5 The full TTITE algorithm

Composing the building blocks of Sections 2.5.2 to 2.5.4 yields the complete TTITE procedure, summarised in Fig. 2.5.

The algorithm is generic in the Hamiltonian: it uses only the Pauli decomposition produced in Section 2.2 and the (small) number of qubits required by that encoding. In this work we apply it to the channel Hamiltonians in the direct encoding (two work qubits plus one ancilla, i.e. three qubits in total).

Remark (Originally on H_2 and Heisenberg models). The algorithm was introduced and demonstrated by Yi et al. [13] on the molecular hydrogen (H_2) Hamiltonian and on one-dimensional Heisenberg spin chains. Because the procedure depends only on the Pauli decomposition of the Hamiltonian, it transfers without modification to the channel Hamiltonians of Eqs. (2.16) to (2.18); we will benchmark its convergence on those Hamiltonians in subsequent chapters.

Algorithm: TTITE.

Input: Pauli decomposition $H = \sum_{i=1}^m c_i h_i$; initial state $|\psi_0\rangle$; total time τ ; step size $\Delta\tau$; Taylor order R .

Output: an approximation to $e^{-H\tau} |\psi_0\rangle / \|e^{-H\tau} |\psi_0\rangle\|$.

1. Set $L = \lceil \tau/\Delta\tau \rceil$ and $|\psi\rangle \leftarrow |\psi_0\rangle$.
2. Repeat L times:
 - (a) For $i = 1, \dots, m$:
 - i. Compute the Taylor coefficients α, β from Eq. (2.64) with $c = c_i$ and $h = h_i$.
 - ii. Build the LCU circuit of Fig. 2.4 with angle Eq. (2.65) and controlled-Pauli string h_i .
 - iii. Execute the circuit on $|0\rangle_{\text{anc}} \otimes |\psi\rangle$; on outcome 0 update $|\psi\rangle$ with the (renormalised) post-selected state; on outcome 1 *restart* the current Trotter step.
3. Return $|\psi\rangle$.

Figure 2.5: Pseudocode for the TTITE algorithm. Each Trotter step requires m successful LCU post-selections; the per-segment success probability is given by Eq. (2.70).

2.5.6 Success probability and resource cost

Each Trotter step requires m LCU circuits to succeed (one per Pauli term), so the segment-level success probability factorises:

$$P_{\text{seg}} = \prod_{i=1}^m P_{s,i}, \quad (2.70)$$

with $P_{s,i}$ given by Eq. (2.69). For small $\Delta\tau$ each individual factor approaches $1/2$, so the per-segment probability decays exponentially in the number of Pauli terms. Two observations soften this otherwise daunting cost.

First, Yi et al. [13] prove that the success probability is *monotonically non-decreasing* in the imaginary-time step: as the state approaches the ground state, applying $H - E_0$ becomes quasi-tangential to it, which inflates the success probabilities $P_{s,i}$. Concretely, with $\tau_\ell = \ell \Delta\tau$ denoting the cumulative imaginary time after the ℓ -th segment,

$$P_{\text{seg}}(\tau_1) \leq P_{\text{seg}}(\tau_2) \leq \dots \leq P_{\text{seg}}(\tau_L), \quad (2.71)$$

so the algorithm becomes *more* efficient as it converges.

Second, the failure probability $1 - P_{s,i}$ can be amplified toward zero by amplitude amplification, at a cost of additional ancillas and depth. We do not employ amplitude amplification in this work, instead relying on the simpler repeat-until-success strategy described in Fig. 2.5. For the two-qubit channel Hamiltonians considered here the per-segment cost

is at most a small multiplicative overhead, and the algorithm remains practical for the imaginary times required to reach the ground state.

The overall picture is therefore the following. TTITE bypasses the parameter-space optimisation of VQE and VQITE by implementing the imaginary-time propagator more or less directly on the device, at the price of an ancilla and a post-selection step per Pauli term. Its accuracy is controlled by two parameters, $\Delta\tau$ and R , and is unaffected by ansatz expressibility. We will see in the next chapters how the three algorithms compare on the same channel Hamiltonians.

2.6 Transition Amplitudes

The algorithms in the previous sections produce eigenstates of the individual channel Hamiltonians. Most experimentally accessible observables, however, involve *matrix elements* of an operator between two such eigenstates—typically a transition operator inducing a radiative decay from a heavier state to a lighter one. In this section we develop the quantum-circuit techniques required to evaluate the two transition types relevant to charmonium spectroscopy: the magnetic-dipole (M1) transitions between spin partners, and the electric-dipole (E1) transitions that change the orbital angular momentum.

2.6.1 Physical motivation

In the multipole expansion of the electromagnetic interaction of a quark–antiquark system, the leading single-photon transitions are classified by their selection rules [10]:

- *M1 transitions* flip the total spin without changing the orbital angular momentum ($\Delta S = 1$, $\Delta L = 0$). In the non-relativistic limit the amplitude reduces to a wavefunction overlap between the two spin partners, scaled by a numerical prefactor.
- *E1 transitions* change the orbital angular momentum by one unit without altering the spin ($\Delta S = 0$, $\Delta L = 1$). The amplitude is proportional to the radial matrix element of the position operator, $\langle\psi_f|r|\psi_i\rangle$.

The two paradigmatic examples for charmonium are $J/\psi \rightarrow \eta_c \gamma$ (M1) and $h_c \rightarrow \eta_c \gamma$ (E1). Computing their amplitudes therefore amounts to evaluating, respectively, the overlap $\langle\psi_f^{(1S_0)}|\psi_i^{(3S_1)}\rangle$ and the matrix element $\langle\psi_f^{(1S_0)}|r|\psi_i^{(1P_1)}\rangle$, where the states are the eigenvectors produced by VQE, VQITE, or TTITE.

2.6.2 M1 transitions: state overlaps

In the non-relativistic limit, the M1 transition amplitude between two states differing only in spin reduces to an overlap of their spatial wavefunctions,

$$|\mathcal{M}_{\text{M1}}|^2 \propto \left| \langle \psi_f^{\text{spatial}} | \psi_i^{\text{spatial}} \rangle \right|^2. \quad (2.72)$$

The two states are eigenvectors of different channel Hamiltonians (1S_0 and 3S_1) but live in the same HO basis, so their overlap is well-defined as a four-dimensional inner product of real coefficient vectors. Three quantum-circuit techniques are available to estimate this overlap.

Inverse-ansatz protocol. If the two ansatz unitaries $U_i = U(\boldsymbol{\theta}_i^*)$ and $U_f = U(\boldsymbol{\theta}_f^*)$ have been determined—e.g., as the output of two VQE or VQITE runs—then the overlap can be read off from the all-zero outcome of the circuit $U_f^\dagger U_i |0\rangle^{\otimes n}$. Indeed,

$$\text{Prob}(\text{all zeros}) = \left| \langle 0 |^{\otimes n} U_f^\dagger U_i | 0 \rangle^{\otimes n} \right|^2 = \left| \langle \psi_f | \psi_i \rangle \right|^2. \quad (2.73)$$

This protocol requires no additional qubits, only an inverse-ansatz sub-circuit, and is the most economical method available when both ansatz unitaries can be efficiently implemented.

Swap test. A more general protocol, applicable when $|\psi_i\rangle$ and $|\psi_f\rangle$ are prepared independently on two copies of the work register, is the swap test [2]. Adding one ancilla initialised in $|0\rangle$, the circuit

$$|0\rangle \otimes |\psi_i\rangle \otimes |\psi_f\rangle \xrightarrow{H_{\text{anc}}} \dots \xrightarrow{\text{CSWAP}} \dots \xrightarrow{H_{\text{anc}}} |\phi\rangle \quad (2.74)$$

yields an ancilla-zero probability $\text{Prob}(0_{\text{anc}}) = \frac{1}{2}(1 + |\langle \psi_f | \psi_i \rangle|^2)$, from which the overlap squared is recovered as $2\text{Prob}(0_{\text{anc}}) - 1$. The swap test is more demanding in qubits ($2n+1$ total) and gates (n controlled-SWAPs), but does not require explicit knowledge of either ansatz unitary, only the ability to prepare both states on demand. The corresponding circuit is shown in Fig. 2.6.

Hadamard test. When the overlap itself rather than its absolute value is required—e.g. for complex amplitudes or for matrix elements of non-unitary operators—the Hadamard test [2] is more appropriate. We discuss it in the context of E1 transitions in the next subsection.

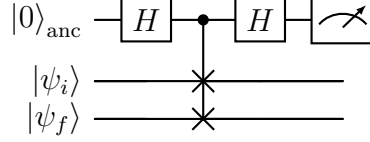


Figure 2.6: Swap test for the absolute overlap between two ansatz states $|\psi_i\rangle$ and $|\psi_f\rangle$. The Hadamard–CSWAP–Hadamard sandwich interferes the swapped and unswapped configurations of the work registers; the ancilla-zero probability is $(1 + |\langle\psi_f|\psi_i\rangle|^2)/2$. The CSWAP gate acts as a controlled exchange of the two work registers.

2.6.3 E1 transitions: matrix elements of the radial operator

For E1 transitions the relevant matrix element is the radial dipole amplitude $\langle\psi_f|r|\psi_i\rangle$, where $|\psi_i\rangle$ and $|\psi_f\rangle$ are eigenstates of the initial- and final-state channels. In the HO basis the radial operator is an $N \times N$ matrix $R_{mk} = \langle\phi_{m\ell_f}|r|\phi_{k\ell_i}\rangle$ which, like the Hamiltonian, can be evaluated in closed form. Using the same techniques that produced Eq. (2.14), one obtains, for the $P \rightarrow S$ case ($\ell_i = 1, \ell_f = 0$) and $N = 4$, the explicit matrix [12]

$$R = \begin{pmatrix} 0.5775 & -0.4715 & 0 & 0 \\ 0 & 0.8821 & -0.6668 & 0 \\ 0 & 0 & 0.8821 & -0.8167 \\ 0 & 0 & 0 & 1.0002 \end{pmatrix} \quad (\text{units of fm}). \quad (2.75)$$

The amplitude is then

$$\langle\psi_f|r|\psi_i\rangle = \mathbf{c}_f^\top R \mathbf{c}_i, \quad (2.76)$$

where $\mathbf{c}_i$ and $\mathbf{c}_f$ are the real coefficient vectors of $|\psi_i\rangle$ and $|\psi_f\rangle$ in the HO basis.

Remark (Errata). Woloshyn’s Eq. (13) reports the (2,2) element of R as 5.4382, which is in fact the numerical value of an entry in the 1S_0 Hamiltonian matrix Eq. (2.16); the correct value, consistent with the monotonically growing radial-element pattern, is 0.8821, as used in Eq. (2.75). This is one of two copy-paste errata noted in Sections 2.1.5 and 2.6.3.

To extract $\langle\psi_f|r|\psi_i\rangle$ on a quantum device, we first Pauli-decompose the operator R on the work register (analogously to Section 2.2.1). The matrix element then reduces to a sum of terms of the form $\langle\psi_f|P|\psi_i\rangle$ for Pauli strings P . Each such term can be estimated by the *Hadamard test*: with one ancilla qubit prepared in $|0\rangle$, the circuit shown in Fig. 2.7 yields an ancilla-zero probability

$$\text{Prob}(0_{\text{anc}}) = \frac{1 + \Re \langle\psi_f|P|\psi_i\rangle}{2}, \quad (2.77)$$

from which the real part of the matrix element is obtained as $2\text{Prob}(0_{\text{anc}}) - 1$. Replacing the leading Hadamard on the ancilla by a $S^\dagger H$ rotation yields, by the same algebra, the imaginary part of the matrix element. For real-symmetric problems such as the channel

Hamiltonians considered here, only the real part is needed.

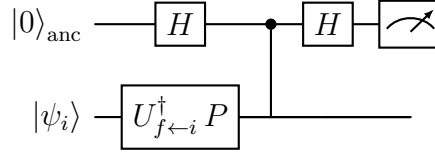


Figure 2.7: Hadamard test for the matrix element $\langle \psi_f | P | \psi_i \rangle$. Conditioned on the ancilla, the work register is rotated by an operator whose expectation value coincides with the desired matrix element. The ancilla-zero probability is $(1 + \Re \langle \psi_f | P | \psi_i \rangle)/2$. Replacing the first Hadamard by $S^\dagger H$ gives the imaginary part.

Putting the pieces together, the E1 transition amplitude is computed as a weighted sum of Hadamard tests, one per Pauli string in the decomposition of R :

$$\langle \psi_f | r | \psi_i \rangle = \sum_k r_k \langle \psi_f | P_k | \psi_i \rangle, \quad (2.78)$$

where r_k are the Pauli coefficients of R obtained from Eq. (2.19). The number of measurements scales with the number of Pauli strings, which for the 4×4 matrix Eq. (2.75) is at most $4^2 = 16$ and typically much smaller after symmetry-based truncation.

With this construction, the same quantum-algorithmic framework that yields energies also yields the matrix elements needed for radiative transitions, allowing a direct comparison with experimental decay widths and lifetimes once eigenstates have been obtained.

2.7 Summary

The framework developed in this chapter equips us with everything needed to run the three algorithms on the charmonium channels of interest. The Hamiltonian construction of Section 2.1 produces three 4×4 real-symmetric matrices encoding the 1S_0 , 3S_1 , and 1P_1 physics; the encodings of Section 2.2 translate each into a sum of Pauli strings on either two or four qubits; the algorithms of Sections 2.3 to 2.5 provide three independent routes to the spectrum of those operators; and the transition machinery of Section 2.6 extends the algorithmic toolkit to observable amplitudes.

The three algorithms differ in significant ways, summarised in Table 2.3. VQE replaces the diagonalisation of the Hamiltonian by a classical optimisation over a parameterised state; its quality is set by the expressibility of the ansatz and by the convergence of the classical optimiser. VQITE retains a parameterised state but evolves it in imaginary time according to the McLachlan principle, trading one optimisation problem for a sequence of small linear systems. TTITE sidesteps parameterisation entirely, implementing the imaginary-time propagator probabilistically through LCU circuits; its cost migrates from classical optimisation to ancilla overhead and post-selection.

Table 2.3: Side-by-side comparison of the three algorithms developed in this chapter, as applied to the channel Hamiltonians of Eqs. (2.16) to (2.18). The “classical workload” refers to the classical computation per iteration; the “quantum workload” refers to the number and depth of circuits executed per iteration. All three algorithms apply to either qubit encoding of Section 2.2.

	VQE	VQITE	TTITE
What is updated	ansatz parameters	ansatz parameters	work-register state itself
What is optimised	energy expectation value	McLachlan residual norm	— (no parameters)
Classical workload	gradient-free optimiser	small linear-system solves	track ancilla outcomes
Quantum workload	$\sim m$ circuits / iter.	$\sim m + p^2$ circuits / step	$\sim m$ LCU circuits / step
Excited states	orthogonality constraints	state-overlap penalties	not directly supported
Ancilla qubits	none	none	one per Trotter step
Dominant error	ansatz expressibility	ansatz + Trotter-like step	Trotter + Taylor + finite shots

A few qualitative consequences of Table 2.3 deserve emphasis. First, VQE and VQITE share the same parameterised ansatz, so their expressibility limits coincide; differences in their numerical performance on the channel Hamiltonians will therefore reflect the relative merits of direct optimisation versus geometry-respecting imaginary-time descent. Second, TTITE achieves accuracy independent of any ansatz, which is attractive in principle but requires an ancilla qubit and at least one controlled-Pauli string per Hamiltonian term, so its hardware cost grows with the size of the Pauli decomposition. Third, transitions, treated in Section 2.6, are post-processing operations on eigenstates produced by any of the three methods; their accuracy is therefore directly inherited from the underlying eigenstate accuracy.

The next chapter applies all three algorithms to the channel Hamiltonians, quantifying these trade-offs through numerical experiments with the codebase developed alongside this thesis.

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